

```
import pandas as pd
from sklearn.datasets import load_iris
iris = load_iris()
```

iris.feature_names

```
['sepal length (cm)',
 'sepal width (cm)',
 'petal length (cm)',
 'petal width (cm)']
```

iris.target_names

```
array(['setosa', 'versicolor', 'virginica'], dtype='<U10')
```

```
df = pd.DataFrame(iris.data, columns=iris.feature_names)
df.head()
```

	sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)	
0	5.1	3.5	1.4	0.2	
1	4.9	3.0	1.4	0.2	
2	4.7	3.2	1.3	0.2	
3	4.6	3.1	1.5	0.2	
4	5.0	3.6	1.4	0.2	

Next steps:

[Generate code with df](#)

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```
df['target'] = iris.target
df.head()
```

	sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)	target	
0	5.1	3.5	1.4	0.2	0	
1	4.9	3.0	1.4	0.2	0	
2	4.7	3.2	1.3	0.2	0	
3	4.6	3.1	1.5	0.2	0	
4	5.0	3.6	1.4	0.2	0	

Next steps:

[Generate code with df](#)

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```
df[df.target==1].head()
```

	sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)	target	
50	7.0	3.2	4.7	1.4	1	
51	6.4	3.2	4.5	1.5	1	
52	6.9	3.1	4.9	1.5	1	
53	5.5	2.3	4.0	1.3	1	
54	6.5	2.8	4.6	1.5	1	

```
df[df.target==2].head()
```

	sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)	target	
100	6.3	3.3	6.0	2.5	2	
101	5.8	2.7	5.1	1.9	2	
102	7.1	3.0	5.9	2.1	2	
103	6.3	2.9	5.6	1.8	2	
104	6.5	3.0	5.8	2.2	2	

```
df['flower_name'] = df.target.apply(lambda x: iris.target_names[x])
df.head()
```

	sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)	target	flower_name	
0	5.1	3.5	1.4	0.2	0	setosa	
1	4.9	3.0	1.4	0.2	0	setosa	
2	4.7	3.2	1.3	0.2	0	setosa	
3	4.6	3.1	1.5	0.2	0	setosa	
4	5.0	3.6	1.4	0.2	0	setosa	

Next steps:

[Generate code with df](#)

[New interactive sheet](#)

```
df[50:101]
```

	sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)	target	flower_name
50	7.0	3.2	4.7	1.4	1	versicolor
51	6.4	3.2	4.5	1.5	1	versicolor
52	6.9	3.1	4.9	1.5	1	versicolor
53	5.5	2.3	4.0	1.3	1	versicolor
54	6.5	2.8	4.6	1.5	1	versicolor
55	5.7	2.8	4.5	1.3	1	versicolor
56	6.3	3.3	4.7	1.6	1	versicolor
57	4.9	2.4	3.3	1.0	1	versicolor
58	6.6	2.9	4.6	1.3	1	versicolor
59	5.2	2.7	3.9	1.4	1	versicolor
60	5.0	2.0	3.5	1.0	1	versicolor
61	5.9	3.0	4.2	1.5	1	versicolor
62	6.0	2.2	4.0	1.0	1	versicolor
63	6.1	2.9	4.7	1.4	1	versicolor
64	5.6	2.9	3.6	1.3	1	versicolor
65	6.7	3.1	4.4	1.4	1	versicolor
66	5.6	3.0	4.5	1.5	1	versicolor
67	5.8	2.7	4.1	1.0	1	versicolor
68	6.2	2.2	4.5	1.5	1	versicolor
69	5.6	2.5	3.9	1.1	1	versicolor
70	5.9	3.2	4.8	1.8	1	versicolor
71	6.1	2.8	4.0	1.3	1	versicolor
72	6.3	2.5	4.9	1.5	1	versicolor
73	6.1	2.8	4.7	1.2	1	versicolor
74	6.4	2.9	4.3	1.3	1	versicolor
75	6.6	3.0	4.4	1.4	1	versicolor
76	6.8	2.8	4.8	1.4	1	versicolor
77	6.7	3.0	5.0	1.7	1	versicolor
78	6.0	2.9	4.5	1.5	1	versicolor
79	5.7	2.6	3.5	1.0	1	versicolor
80	5.5	2.4	3.8	1.1	1	versicolor
81	5.5	2.4	3.7	1.0	1	versicolor
82	5.8	2.7	3.9	1.2	1	versicolor
83	6.0	2.7	5.1	1.6	1	versicolor
84	5.4	3.0	4.5	1.5	1	versicolor
85	6.0	3.4	4.5	1.6	1	versicolor
86	6.7	3.1	4.7	1.5	1	versicolor
87	6.3	2.3	4.4	1.3	1	versicolor
88	5.6	3.0	4.1	1.3	1	versicolor
89	5.5	2.5	4.0	1.3	1	versicolor
90	5.5	2.6	4.4	1.2	1	versicolor
91	6.1	3.0	4.6	1.4	1	versicolor
92	5.8	2.6	4.0	1.2	1	versicolor
93	5.0	2.3	3.3	1.0	1	versicolor
94	5.6	2.7	4.2	1.3	1	versicolor
95	5.7	3.0	4.2	1.2	1	versicolor
96	5.7	2.9	4.2	1.3	1	versicolor
97	6.2	2.9	4.3	1.3	1	versicolor
98	5.1	2.5	3.0	1.1	1	versicolor
99	5.7	2.8	4.1	1.3	1	versicolor
100	6.3	3.3	6.0	2.5	2	virginica

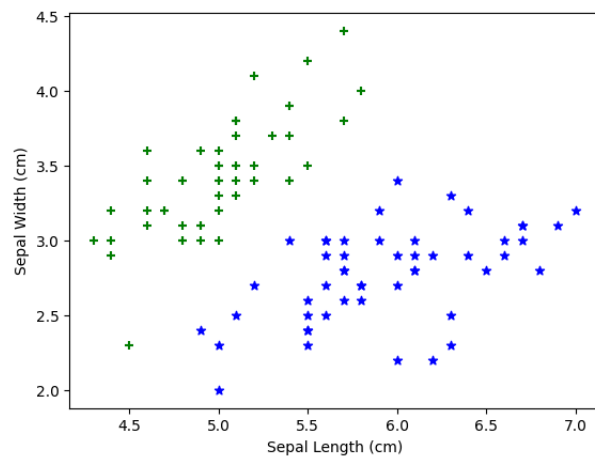
```
df0=df[:50]
df1= df[50:100]
df2 =df[100:]
```

```
import matplotlib.pyplot as plt
%matplotlib inline
```

```
plt.xlabel('Sepal Length (cm)')
plt.ylabel('Sepal Width (cm)')

plt.scatter(df0['sepal length (cm)'], df0['sepal width (cm)'], color="green", marker='+')
plt.scatter(df1['sepal length (cm)'], df1['sepal width (cm)'], color="blue", marker='*')

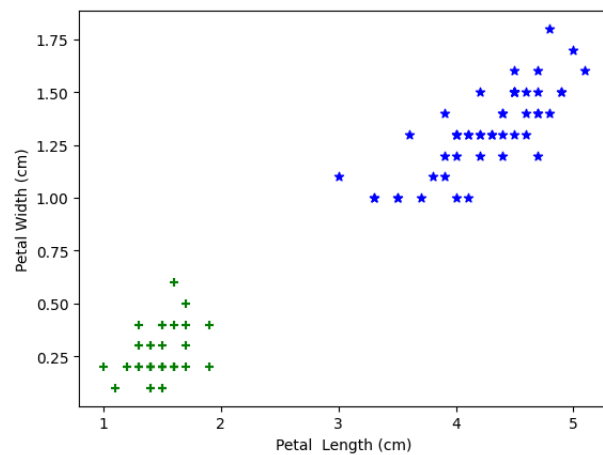
plt.show()
```



```
plt.xlabel('Petal Length (cm)')
plt.ylabel('Petal Width (cm)')

plt.scatter(df0['petal length (cm)'], df0['petal width (cm)'], color="green", marker='+')
plt.scatter(df1['petal length (cm)'], df1['petal width (cm)'], color="blue", marker='*')

plt.show()
```



```
from sklearn.model_selection import train_test_split
```

```
X = df.drop(['target', 'flower_name'], axis = 'columns')
y = df.target
```

```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3)
```

```
len(X_train)
```

```
105
```

```
len(X_test)
```

```
45
```

```
from sklearn.svm import SVC
model = SVC()
```

```
model.fit(X_train, y_train)
```

```
▼ SVC ⓘ ?
SVC()
```

```
model.score(X_test, y_test)
```

```
0.9777777777777777
```

```
model.predict([[4.8, 3.0, 1.5, 0.3]])
```

```
/usr/local/lib/python3.12/dist-packages/sklearn/utils/validation.py:2739: UserWarning: X does not have valid feature names, but SVC was fitted with feature names
warnings.warn(
array([0])
```

<pre>model_C = SVC(C=1) model_C.fit(X_train,y_train) model_C.score(X_test,y_test)</pre>	
0.9777777777777777	
<pre>model_C = SVC(C=10) model_C.fit(X_train,y_train) model_C.score(X_test,y_test)</pre>	
0.9777777777777777	
<pre>model_g = SVC(gamma=10) model_g.fit(X_train,y_train) model_g.score(X_test,y_test)</pre>	
0.8888888888888888	
<pre>model_linear_kernal = SVC(kernel = 'linear') model_linear_kernal.fit(X_test,y_test)</pre>	
▼ SVC ⓘ ? SVC(kernel='linear')	
<pre>model_linear_kernal.score(X_test, y_test)</pre>	
1.0	